TikuKits ML: Linear Regression for Embedded Systems

Simple. Ubiquitous. Intelligence, Everywhere.

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Outline

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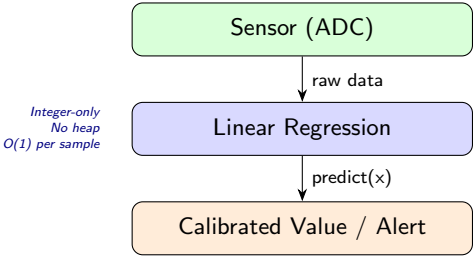
Why ML on Ultra-Low-Power MCUs?

Embedded Use Cases

- **Sensor calibration** — map raw ADC to physical units
- **Trend detection** — detect drift in readings over time
- **Prediction** — estimate battery charge from voltage
- **Anomaly flagging** — R^2 drop signals sensor failure

Constraints

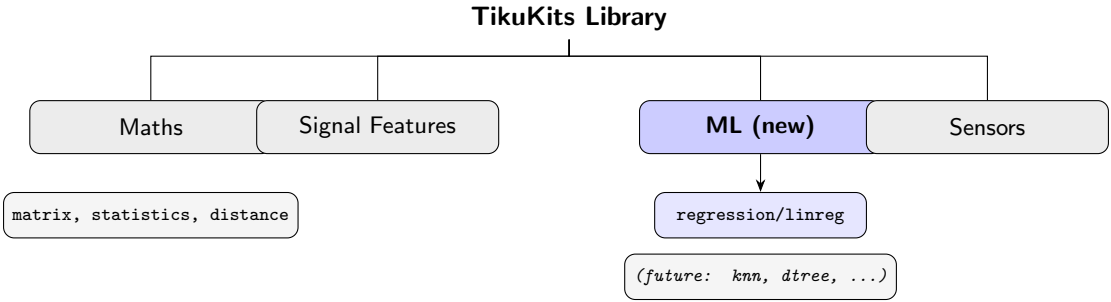
- No FPU (16-bit MSP430)
- No heap allocation
- Limited RAM (2–8 KB)
- Every µJ counts



Design Goal

Bring linear regression to devices with **no floating point**, using only static memory and integer arithmetic.

ML Kit in the TikuKits Ecosystem



File	Role
`ml/tiku_kits_ml.h`	Common return codes for all ML sub-modules
`ml/regression/tiku_kits_ml_linreg.h`	Linear regression API
`ml/regression/tiku_kits_ml_linreg.c`	Implementation

Ordinary Least Squares (OLS)

Model: $y = \beta_1 x + \beta_0$

Given n data points (x_i, y_i), the OLS solution minimises $\sum (y_i - \hat{y}_i)^2$:

$$\beta_1 = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - \left(\sum x_i\right)^2}$$

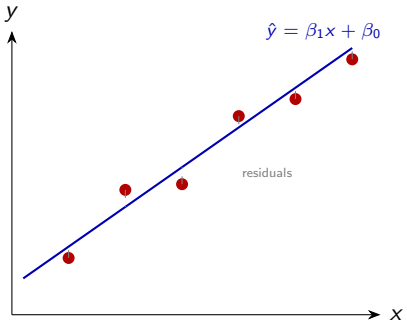
$$\beta_0 = \frac{\sum y_i - \beta_1 \sum x_i}{n}$$

Prediction:

$$\hat{y} = \beta_1 x + \beta_0$$

Goodness of fit:

$$R^2 = \frac{\left(nS_{xy} - S_x S_y\right)^2}{\left(nS_{xx} - S_x^2\right)\left(nS_{yy} - S_y^2\right)}$$



Key Insight

OLS only needs **five running sums**:
S_x, S_y, S_{xx}, S_{xy}, S_{yy}.
No sample buffer required — O(1) memory.

Integer-Only Implementation Strategy

Problem

Slope and intercept are typically **fractional**. MSP430 has no FPU — division truncates to zero.

$\beta_1 = 1.5 \rightarrow$ `int` gives 1 (33% error).

Solution: Fixed-Point Q-Format

Scale by 2^{shift} before dividing:

$$\text{slope_q} = \frac{\text{num} \ll \text{shift}}{\text{den}}$$

With `shift=8`, resolution $= \frac{1}{256} \approx 0.004$.

To recover real value: slope $= \frac{\text{slope_q}}{256}$.

Overflow Prevention

- All accumulators use `int64_t`
- Products $x \cdot y$ computed as `(int64_t)x * (int64_t)y`
- Safe for `int32_t` inputs: $2^{31} \times 2^{31} = 2^{62} < 2^{63}$

Prediction Without Scaling

`predict()` returns values in the **original integer domain** — no Q-format decoding needed:

$$\hat{y} = \frac{\text{num} \cdot (n \cdot x - S_x) + S_y \cdot \text{den}}{n \cdot \text{den}}$$

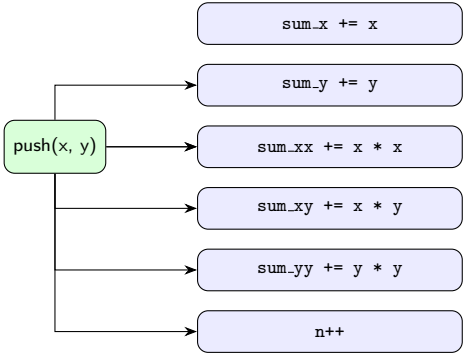
The tiku_kits_ml_linreg Structure

ml/regression/tiku_kits_ml_linreg.h

```
struct tiku_kits_ml_linreg {
    int64_t sum_x;    /* Sum of x        */
    int64_t sum_y;    /* Sum of y        */
    int64_t sum_xx;   /* Sum of x*x      */
    int64_t sum_xy;   /* Sum of x*y      */
    int64_t sum_yy;   /* Sum of y*y      */
    uint16_t n;       /* Data points     */
    uint8_t shift;    /* Q-format bits   */
};
```

Memory footprint:
$5 \times 8 + 2 + 1 = 43$ bytes per model.
All statically allocated — no heap.

Streaming accumulator pattern:



Each `push()` is **O(1)** — five multiply-add operations.

ML Kit Return Codes

ml/tiku_kits_ml.h

```
#define TIKU_KITS_ML_OK            0
#define TIKU_KITS_ML_ERR_NULL     (-1)
#define TIKU_KITS_ML_ERR_SIZE     (-2)
#define TIKU_KITS_ML_ERR_PARAM    (-3)
#define TIKU_KITS_ML_ERR_SINGULAR (-4)
#define TIKU_KITS_ML_ERR_OVERFLOW (-5)
```

Defined in the **kit-level common header** — shared by all ML sub-modules (linear regression, and future modules like kNN, decision tree, etc.).

Code	Meaning
OK	Operation succeeded
ERR_NULL	NULL pointer passed
ERR_SIZE	Fewer than 2 data points
ERR_PARAM	Invalid parameter (e.g. shift ¿ 30)
ERR_SINGULAR	All x values identical (zero variance)
ERR_OVERFLOW	Arithmetic overflow

Complete API

Function	Description
`tiku_kits_ml_linreg_init(lr, shift)`	Initialize model with Q-format precision (shift = 0..30)
`tiku_kits_ml_linreg_reset(lr)`	Clear all data points, preserve shift
`tiku_kits_ml_linreg_push(lr, x, y)`	Stream a data point — O(1), 5 multiply-adds
`tiku_kits_ml_linreg_slope(lr, &result)`	Fitted slope $\times 2^{\text{shift}}$ (Q-format)
`tiku_kits_ml_linreg_intercept(lr, &result)`	Fitted intercept $\times 2^{\text{shift}}$ (Q-format)
`tiku_kits_ml_linreg_predict(lr, x, &result)`	Predicted $\hat{y}$ in original integer scale
`tiku_kits_ml_linreg_r2(lr, &result)`	$R^2 \times 2^{\text{shift}}$; range $[0, 2^{\text{shift}}]$
`tiku_kits_ml_linreg_count(lr)`	Number of data points pushed

Minimum Data Requirement

`slope()`, `intercept()`, `predict()`, and `r2()` require ≥ 2 data points with distinct x values. Returns `ERR_SIZE` or `ERR_SINGULAR` otherwise.

Compile-Time Configuration

Configurable element type

```c
/* Default: int32_t (no FPU needed) */
#ifndef TIKU_KITS_ML_ELEM_TYPE
#define TIKU_KITS_ML_ELEM_TYPE int32_t
#endif

typedef TIKU_KITS_ML_ELEM_TYPE
    tiku_kits_ml_elem_t;

/* Override for tight memory: */
#define TIKU_KITS_ML_ELEM_TYPE int16_t
#include "tiku_kits_ml_linreg.h"
```

Configurable Q-format precision

```c
/* Default: 8 fractional bits */
#ifndef TIKU_KITS_ML_LINREG_SHIFT
#define TIKU_KITS_ML_LINREG_SHIFT 8
#endif

/* Or set at runtime: */
tiku_kits_ml_linreg_init(&lr, 12);
/* 12-bit: resolution ~0.00024 */
```

Shift	Scale	Resolution
4	16	0.0625
8	256	0.0039
12	4096	0.00024
16	65536	0.000015

Core: Push & Slope

tiku_kits_ml_linreg.c — push

```c
int tiku_kits_ml_linreg_push(
    struct tiku_kits_ml_linreg *lr,
    tiku_kits_ml_elem_t x,
    tiku_kits_ml_elem_t y)
{
    if (lr == NULL)
        return TIKU_KITS_ML_ERR_NULL;

    lr->sum_x  += (int64_t)x;
    lr->sum_y  += (int64_t)y;
    lr->sum_xx += (int64_t)x * (int64_t)x;
    lr->sum_xy += (int64_t)x * (int64_t)y;
    lr->sum_yy += (int64_t)y * (int64_t)y;
    lr->n++;

    return TIKU_KITS_ML_OK;
}
```

tiku_kits_ml_linreg.c — slope

```c
int tiku_kits_ml_linreg_slope(
    const struct tiku_kits_ml_linreg *lr,
    int32_t *result)
{
    int64_t num, den;
    int rc;

    if (lr == NULL || result == NULL)
        return TIKU_KITS_ML_ERR_NULL;

    rc = slope_parts(lr, &num, &den);
    if (rc != TIKU_KITS_ML_OK)
        return rc;

    /* Fixed-point scaling */
    *result = (int32_t)(
        (num << lr->shift) / den);
    return TIKU_KITS_ML_OK;
}
```

Core: Predict & R-Squared

predict — returns original scale

```c
int tiku_kits_ml_linreg_predict(
    const struct tiku_kits_ml_linreg *lr,
    tiku_kits_ml_elem_t x,
    tiku_kits_ml_elem_t *result)
{
    int64_t num, den, y_num;
    /* ... NULL/size checks ... */

    /*
     * y = (num*(n*x - Sx) + Sy*den)
     *     / (n * den)
     *
     * No fixed-point decoding needed.
     */
    y_num = num * ((int64_t)lr->n
                 * (int64_t)x - lr->sum_x)
            + lr->sum_y * den;

    *result = (tiku_kits_ml_elem_t)(
        y_num / ((int64_t)lr->n * den));
    return TIKU_KITS_ML_OK;
}
```

R-squared — goodness of fit

```c
int tiku_kits_ml_linreg_r2(
    const struct tiku_kits_ml_linreg *lr,
    int32_t *result)
{
    int64_t num, den, ss_yy;
    /* ... NULL/size checks ... */

    /* SS_yy = n*Syy - Sy^2 */
    ss_yy = (int64_t)lr->n * lr->sum_yy
            - lr->sum_y * lr->sum_y;

    if (ss_yy == 0)
        return TIKU_KITS_ML_ERR_SINGULAR;

    /* R^2 = num^2 / (den * ss_yy)
     * scaled by (1 << shift)        */
    *result = (int32_t)(
        ((num / den) * num << lr->shift)
        / ss_yy);

    /* Clamp to [0, 1<<shift] */
    if (*result > (int32_t)(1<<lr->shift))
        *result = (int32_t)(1<<lr->shift);
    if (*result < 0) *result = 0;
```

Internal Helper: Slope Decomposition

slope_parts — shared by slope, intercept, predict, r2

```c
static int slope_parts(const struct tiku_kits_ml_linreg *lr,
                       int64_t *num, int64_t *den)
{
    int64_t d;

    if (lr->n < 2)
        return TIKU_KITS_ML_ERR_SIZE;

    *num = (int64_t)lr->n * lr->sum_xy - lr->sum_x * lr->sum_y;

    d = (int64_t)lr->n * lr->sum_xx - lr->sum_x * lr->sum_x;
    if (d == 0)
        return TIKU_KITS_ML_ERR_SINGULAR;    /* all x values identical */

    *den = d;
    return TIKU_KITS_ML_OK;
}
```

Why a Helper?

All four query functions (`slope`, `intercept`, `predict`, `r2`) need the same numerator and denominator decomposition. Factoring it out avoids code duplication and ensures consistent error handling.

Example 1: Sensor Calibration

examples/example_ml_linreg.c — ADC to temperature

```c
struct tiku_kits_ml_linreg lr;
int32_t slope_q, intercept_q, r2_q;
tiku_kits_ml_elem_t y_pred;

tiku_kits_ml_linreg_init(&lr, 8);

/* Calibration data: (raw_adc, temperature_deci_C) */
tiku_kits_ml_linreg_push(&lr, 100, 0);      /* 0.0 C */
tiku_kits_ml_linreg_push(&lr, 200, 100);    /* 10.0 C */
tiku_kits_ml_linreg_push(&lr, 300, 200);    /* 20.0 C */
tiku_kits_ml_linreg_push(&lr, 400, 300);    /* 30.0 C */
tiku_kits_ml_linreg_push(&lr, 500, 400);    /* 40.0 C */

tiku_kits_ml_linreg_slope(&lr, &slope_q);        /* 256 (1.0 Q8) */
tiku_kits_ml_linreg_intercept(&lr, &intercept_q); /* -25600 (-100.0 Q8) */
tiku_kits_ml_linreg_r2(&lr, &r2_q);              /* 256 (1.0 Q8, perfect) */

/* Convert a new ADC reading to temperature */
tiku_kits_ml_linreg_predict(&lr, 350, &y_pred);  /* y_pred = 250 (25.0 C) */
```

Example 2: Trend Detection

Detect drift in sensor readings

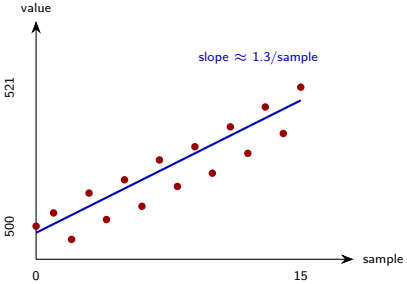
```
struct tiku_kits_ml_linreg lr;
int32_t slope_q;

static const tiku_kits_ml_elem_t
    readings[] = {
    500, 502, 498, 505, 501, 507,
    503, 510, 506, 512, 508, 515,
    511, 518, 514, 521
};

tiku_kits_ml_linreg_init(&lr, 8);
for (i = 0; i < 16; i++)
    tiku_kits_ml_linreg_push(&lr,
        (tiku_kits_ml_elem_t)i,
        readings[i]);

tiku_kits_ml_linreg_slope(&lr, &slope_q);

if (slope_q > 25)        /* > ~0.1/sample */
    alert_upward_drift();
else if (slope_q < -25)
    alert_downward_drift();
```



Use Case

Detects gradual sensor drift (e.g. temperature creep, battery discharge) from a stream of readings. A slope near zero = stable; non-zero = drifting.

Example 3: Battery Charge Prediction

Map battery voltage (mV) to charge level (percent $\times$ 10)

```
struct tiku_kits_ml_linreg lr;
tiku_kits_ml_elem_t y_pred;

tiku_kits_ml_linreg_init(&lr, 8);

/* Calibration: voltage -> charge*10 */
tiku_kits_ml_linreg_push(&lr, 3000, 0);      /* 0% */
tiku_kits_ml_linreg_push(&lr, 3300, 250);    /* 25% */
tiku_kits_ml_linreg_push(&lr, 3600, 500);    /* 50% */
tiku_kits_ml_linreg_push(&lr, 3900, 750);    /* 75% */
tiku_kits_ml_linreg_push(&lr, 4200, 1000);   /* 100% */

/* Predict charge for measured voltage */
tiku_kits_ml_linreg_predict(&lr, 3450, &y_pred);  /* 375 -> 37.5% */
tiku_kits_ml_linreg_predict(&lr, 3750, &y_pred);  /* 625 -> 62.5% */
```

Practical Note

Real battery discharge curves are non-linear. Linear regression provides a reasonable **first approximation** for the near-linear mid-range (20–80%) and is sufficient for many embedded applications where accuracy within 5% is acceptable.

Test Cases Overview

#	Test Function	What It Verifies
1	`test_kits_ml_linreg_perfect_fit`	$y = 2x$: slope = 512 (2.0 Q8), intercept = 0
2	`test_kits_ml_linreg_intercept`	$y = 2x + 1$: slope = 512, intercept = 256 (1.0 Q8)
3	`test_kits_ml_linreg_fractional_slope`	$y = 1.5x + 10$: slope = 384 (1.5 Q8), intercept = 2560
4	`test_kits_ml_linreg_predict`	Prediction at training, interpolation, and extrapolation points
5	`test_kits_ml_linreg_r2`	Perfect fit $\rightarrow R^2 = 256$; noisy data $\rightarrow 0 < R^2 < 256$
6	`test_kits_ml_linreg_negative_values`	$y = -2x + 100$: negative slope and large intercept
7	`test_kits_ml_linreg_reset`	Reset clears data; refit produces correct new model
8	`test_kits_ml_linreg_edge_cases`	1 point $\rightarrow$ ERR_SIZE; identical $x \rightarrow$ SINGULAR; constant y
9	`test_kits_ml_linreg_null_inputs`	All functions reject NULL pointers with ERR_NULL

Test Deep Dive: Perfect Fit & Edge Cases

Perfect fit: $y = 2x$

```c
void test_kits_ml_linreg_perfect_fit(void)
{
    struct tiku_kits_ml_linreg lr;
    int32_t slope_q;

    TEST_GROUP_BEGIN("LinReg Perfect Fit");

    tiku_kits_ml_linreg_init(&lr, 8);
    tiku_kits_ml_linreg_push(&lr, 1, 2);
    tiku_kits_ml_linreg_push(&lr, 2, 4);
    tiku_kits_ml_linreg_push(&lr, 3, 6);

    tiku_kits_ml_linreg_slope(
        &lr, &slope_q);
    TEST_ASSERT(slope_q == 512,
        "slope(y=2x) == 512 (2.0 Q8)");

    TEST_GROUP_END("LinReg Perfect Fit");
}
```

Edge cases: singular data

```c
void test_kits_ml_linreg_edge_cases(void)
{
    struct tiku_kits_ml_linreg lr;
    int32_t slope_q;
    int rc;

    TEST_GROUP_BEGIN("LinReg Edge Cases");

    /* Single point: too few */
    tiku_kits_ml_linreg_init(&lr, 8);
    tiku_kits_ml_linreg_push(&lr, 5, 10);
    rc = tiku_kits_ml_linreg_slope(
        &lr, &slope_q);
    TEST_ASSERT(
        rc == TIKU_KITS_ML_ERR_SIZE,
        "slope with 1 point rejected");

    /* Identical x: singular */
    tiku_kits_ml_linreg_reset(&lr);
    tiku_kits_ml_linreg_push(&lr, 5, 10);
    tiku_kits_ml_linreg_push(&lr, 5, 20);
    rc = tiku_kits_ml_linreg_slope(
        &lr, &slope_q);
    TEST_ASSERT(
```

ML Kit Files

Layer	File	Role
Common	`tikukits/ml/tiku_kits_ml.h`	Shared return codes for all ML sul modules
Linear Regression	`tikukits/ml/regression/tiku_kits_ml_linreg.h` `tikukits/ml/regression/tiku_kits_ml_linreg.c`	API, struct, configuration macros Implementation (43 bytes state, O(1 push)
Tests	`tikukits/tests/ml/test_kits_ml.h` `tikukits/tests/ml/test_kits_ml_linreg.c`	Test function declarations 9 test groups, 22+ assertions
Examples	`tikukits/examples/example_ml_linreg.c`	3 practical examples (calibration, trend prediction)

Convention for New ML Sub-Modules

1. Place files in `ml/<category>/tiku_kits_ml_<module>.h/.c`
2. Include `../tiku_kits_ml.h` for shared return codes

Summary

What We Built

- Streaming OLS linear regression
- Integer-only / fixed-point arithmetic
- 43 bytes of state, no heap
- O(1) per sample push
- On-demand slope, intercept, predict, R^2

Key Design Choices

- `int64_t` accumulators prevent overflow
- Configurable element type and Q-shift
- `predict()` returns original-scale values
- Follows TikuKits naming and structure conventions

Use Cases

- Sensor calibration (ADC $\rightarrow$ physical units)
- Trend detection (drift monitoring)
- Battery charge estimation
- Anomaly flagging via R^2 monitoring

Test Coverage

9 test groups covering: perfect fit, non-zero intercept, fractional slope, prediction, R^2, negative values, reset, edge cases, and NULL safety.

Future ML Sub-Modules

k-NN classifier, decision tree, perceptron — same

Thank You

TikuKits ML: Linear Regression

Integer-Only Machine Learning for Physical AI Endpoints

Repository:	`https://github.com/tikuos/tikuOS`
License:	Apache 2.0
Common:	`tikukits/ml/tiku_kits_ml.h`
Header:	`tikukits/ml/regression/tiku_kits_ml_linreg.h`
Source:	`tikukits/ml/regression/tiku_kits_ml_linreg.c`
Tests:	`tikukits/tests/ml/test_kits_ml_linreg.c`
Examples:	`tikukits/examples/example_ml_linreg.c`